# FoodGuard — Packaged Food Safety & Regulatory Assessment System

Standardized 300-Record Food Additive, Adulteration & Multi-Country Regulatory Dataset Specification

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| **Project Domain:** | Food Safety, Regulatory Compliance & Computer Vision OCR | **Dataset File:** | 300_packaged_food_dataset.csv |
| **Total Records:** | 300 Standardized Product/Additive Records | **Authorities:** | India (FSSAI), USA (FDA), EU (EFSA), Australia (FSANZ) |
| **Architecture:** | React + FastAPI + Python Decision Engine + PaddleOCR | **Date:** | Academic Year 2026 |

## 1. Executive Summary & Project Purpose

Modern packaged food supply chains contain complex combinations of synthetic dyes, chemical preservatives, stabilizers, sweeteners, and texturizers. Crucially, permissible usage levels and legal authorizations diverge widely across international food regulatory bodies. An additive legally permitted in one country may be strictly banned or revoked in another due to emerging toxicological or carcinogenicity evidence (e.g. Titanium Dioxide INS 171, Erythrosine INS 127, Potassium Bromate INS 924a, Azodicarbonamide INS 927a).

The **FoodGuard 300-Record Regulatory Dataset** provides an empirical, structured ground-truth benchmark enabling automated cross-referencing of packaged food ingredients extracted via **1D/2D Barcode Scanning** and **PaddleOCR Vision Analysis** against statutory permissions in India, the United States, the European Union, and Australia.

## 2. Standardized 15-Column Dataset Schema

The dataset is structured as a normalized CSV file (`300_packaged_food_dataset.csv`) with 15 standard attributes:

| Column Name | Type | Description & Example |
|---|---|---|
| **record_id** | Integer | Unique sequential identifier (1 to 300). |
| **brand_or_manufacturer** | String | Target manufacturer or category brand (Parle, Britannia, Nestlé, PepsiCo, etc.). |
| **product_category** | String | Food sector (Beverage, Confectionery, Extruded Snack, Bakery, Dairy, Spices). |
| **product_name** | String | Packaged item or additive formulation name. |
| **common_label_name** | String | Consumer-facing ingredient list name (e.g. Tartrazine, Sodium Benzoate, MSG). |
| **scientific_chemical_name** | String | IUPAC chemical nomenclature (e.g. Disodium 2,4,5,7-tetraiodofluorescein). |
| **codex_ins_no** | String | Codex Alimentarius International Numbering System code (e.g. 102, 171, 924a). |
| **fssai_reference_status** | String | Statutory status in India under FSSAI Food Additives Regulations. |
| **usa_reference_status** | String | Statutory status in USA under US FDA Title 21 Code of Federal Regulations. |
| **eu_reference_status** | String | Statutory status in European Union under Regulation (EC) No 1333/2008. |
| **eu_e_number** | String | EU authorized E-number code (e.g. E102, E127, E171, E927a). |
| **us_identifier_reference** | String | FDA CFR regulatory section (e.g. 21 CFR 74.705, 21 CFR 184.1733). |
| **australia_nz_code** | String | Food Standards Australia New Zealand (FSANZ) Code identifier. |
| **screening_note** | String | Toxicological reason, mandatory label warning, or category limit instruction. |
| **source** | String | Ground truth provenance citation note. |

## 3. Cross-Country Regulatory Compliance Matrix

The core functionality of FoodGuard is detecting divergent legal authorizations between global food unions. The table below illustrates sample high-impact regulatory divergences extracted directly from the dataset:

| INS / Name | India (FSSAI) | USA (US FDA) | EU (EFSA) | Regulatory Screening Rationale |
|---|---|---|---|---|
| **INS 171**<br>Titanium Dioxide | Specified applications | Permitted (21 CFR 73.575) | **Not authorised / Banned** | Banned across EU under Regulation (EU) 2022/63 due to DNA strand break concerns. |
| **INS 127**<br>Erythrosine (Red 3) | Permitted with limits | **Revoked / Phase-out** | **Restricted to cherries** | FDA revoked color authorization under 21 CFR 74.303; EU restricted due to thyroid tumorigenesis. |
| **INS 924a**<br>Potassium Bromate | **Not permitted (Banned)** | Permitted in bakery | **Not authorised / Banned** | Banned by FSSAI (2016) and EU as an IARC Group 2B probable human carcinogen. |
| **INS 927a**<br>Azodicarbonamide | Specified flour uses | Permitted (21 CFR 172.806) | **Not authorised / Banned** | Banned in Europe and UK due to respiratory sensitization and semicarbazide breakdown. |
| **BVO**<br>Brominated Veg. Oil | Category verification | **Revoked / Banned** | **Not authorised / Banned** | FDA finalized rule revoking BVO authorization in 2024 following bioaccumulation studies. |
| **INS 143**<br>Fast Green FCF | Specified uses | Permitted | **Not authorised in EU** | Not approved under EU food additive registry; permitted in US with batch certification. |
| **INS 952**<br>Cyclamate | Category verification | **Not permitted (Banned)** | Permitted with limits | Strictly prohibited as food sweetener in US since 1970; widely permitted in EU and UK. |
| **INS 102**<br>Tartrazine (Yellow 5) | Permitted with limits | Permitted (21 CFR 74.705) | Warning label required | Mandatory EU warning: 'May have an adverse effect on activity and attention in children'. |

## 4. Algorithmic Matching & Regulatory Audit Flow

When an input is received (via 1D Barcode scanning, manual barcode entry, or PaddleOCR label photo extraction), the FoodGuard decision engine executes the following verification pipeline:

- **Step 1 — Entity Normalization:** Isolates additive names and Codex INS codes using regular expressions (`r'\bins\s*(\d+)'`, `r'\be\s*(\d+)'`).
- **Step 2 — Dataset Cross-Referencing:** Matches extracted tokens against indexed dataset records in `300_packaged_food_dataset.csv`.
- **Step 3 — Multi-Country Permission Check:** Queries statutory compliance across India (FSSAI), USA (FDA), EU (EFSA), and Australia (FSANZ).
- **Step 4 — Decision Synthesis:**

  a) If *all* permissions are Permitted: System displays clean status (✓ **All Permissions OK — Low Risk**).

  b) If *any* authority revokes or bans the ingredient: System generates an explicit alert specifying the **INS Number**, **Item Name**, **Banning Authority**, and **Toxicological Rationale**.

## 5. Dataset Statistical Distribution

| Category / Segment | Count | Representative Substance Examples |
|---|---|---|
| **Food Colours & Pigments** | 68 | Tartrazine, Sunset Yellow, Allura Red, Brilliant Blue, Erythrosine, Titanium Dioxide |
| **Preservatives & Antioxidants** | 74 | Sodium Benzoate, Potassium Sorbate, BHA, BHT, TBHQ, Sodium Metabisulphite |
| **Emulsifiers, Stabilizers & Gums** | 52 | Lecithin, PGPR, Guar Gum, Xanthan Gum, DATEM, Mono- and Diglycerides |
| **Artificial & Natural Sweeteners** | 38 | Aspartame, Acesulfame-K, Sucralose, Saccharin, Cyclamate, Stevia, Xylitol |
| **Flavour Enhancers & Acidulants** | 36 | MSG (INS 621), Disodium Inosinate (INS 631), Guanylate (INS 627), Citric Acid |
| **Banned Chemical Adulterants** | 32 | Metanil Yellow, Lead Chromate, Rhodamine B, Sudan Dyes, Melamine, Formalin |
| **Total Standardized Records** | **300** | **Full empirical coverage across 10 packaged food FMCG sectors** |

## 6. Academic Verification & Project Mentor Sign-Off

This dataset and verification system have been designed and validated in accordance with the regulatory statutes of the Food Safety and Standards Authority of India (FSSAI), the United States Code of Federal Regulations (21 CFR), and the European Food Safety Authority (EFSA Regulation 1333/2008).

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| **Submitted by:** | Project Student / Team | **Date of Submission:** | October 2026 |
| **Verified by:** | Project Mentor / Guide | **Mentor Signature:** | _____ |